

Clinical Training Handbook

A Narrative Guide to Condition-Specific Pathways and Management

Prepared from Clinical_Guidelines_Compact.docx.pdf

This handbook is written for clinician training. It presents each condition in natural language, with clear pathway selection logic and practical management orientation at the bedside.

Condition 1: Abnormal Uterine Bleeding in Non

If the patient is haemodynamically stable but actively bleeding, start intravenous (IV) crystalloid while arranging admission and urgent gynaecological assessment. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

Pathway differentiation is driven by explicit trigger statements in the source guideline. Key branch rules include: If the patient is haemodynamically stable but actively bleeding, start intravenous (IV) crystalloid while arranging admission and urgent gynaecological assessment. If bleeding has stopped and there are no red flags—such as thrombocytopenia, symptomatic anaemia (haemoglobin below 10 grams per decilitre), suspected coagulopathy, or recurrent heavy bleeding—outpatient review may be considered only with clear return precautions and confirmed follow-up within 24 hours. When granulocytosis is present, consider infection or inflammatory disease and add liver function tests if systemic illness is suspected. When thrombocytopenia is present, request a coagulation profile and peripheral blood film, and consider immune thrombocytopenic purpura, von Willebrand disease, disseminated intravascular coagulation, drug-related thrombocytopenia, or haematological malignancy. These branch statements should be interpreted sequentially, with immediate stabilization pathways prioritized before definitive or downstream treatment pathways.

Management should be delivered as staged care: immediate stabilization, targeted diagnostics, condition-specific therapeutic intervention, and planned reassessment. For training purposes, each decision point should be documented with the rationale for branch selection, the intervention chosen, and the expected response window that would prompt either escalation or de-escalation of care.

In follow-up discussions, trainees should be taught to ask one focused clarifying question when pathway selection is uncertain, rather than expanding broad differentials without direction. This preserves decisional clarity while maintaining safety in ambiguous presentations.

Condition 2: Pregnant Patient Under 20 Weeks Gestation When a pregnant patient presents with per vaginal bleeding before 20 weeks' gestation, manage as threatened, incomplete, inevitable, missed, septic, or complete abortion until ectopic pregnancy has been excluded by obstetric ultrasound and serum human chorionic gonadotropin (h trend

If the patient is shocked, initiate immediate airway, breathing, and circulation support using the ABCDE approach; secure intravenous access; administer 1 litre of 0.9% sodium chloride over 15 minutes; send urgent complete blood count, blood group and cross-match, renal function and electrolytes, and arrange urgent obstetric ultrasound. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological

stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

Pathway differentiation is driven by explicit trigger statements in the source guideline. Key branch rules include: If the patient is shocked, initiate immediate airway, breathing, and circulation support using the ABCDE approach; secure intravenous access; administer 1 litre of 0.9% sodium chloride over 15 minutes; send urgent complete blood count, blood group and cross-match, renal function and electrolytes, and arrange urgent obstetric ultrasound. When obstetric ultrasound confirms retained products of conception and the patient is haemodynamically stable, medical or surgical evacuation should be performed under gynaecological supervision. When granulocytosis is present alongside fever, uterine tenderness, or offensive vaginal discharge, suspect septic abortion and administer intravenous ceftriaxone and metronidazole while arranging evacuation. When thrombocytopenia or coagulopathy is identified, urgently assess for disseminated intravascular coagulation and involve haematology and obstetric specialists. These branch statements should be interpreted sequentially, with immediate stabilization pathways prioritized before definitive or downstream treatment pathways.

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Condition 3: Pregnant Patient At or Beyond 20 Weeks Gestation When a pregnant patient at or beyond 20 weeks' gestation presents with painless per vaginal bleeding, placenta previa must be suspected until excluded by transabdominal and transvaginal obstetric ultrasound

If placenta previa is confirmed and bleeding is active or recurrent, admit immediately, restrict vaginal examination, avoid rectal interventions, monitor fetal heart rate continuously, administer intravenous tranexamic acid within 3 hours of presentation, and prepare for emergency delivery if maternal or fetal compromise develops. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

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Condition 4: Postpartum Haemorrhage When a patient bleeds more than 500 millilitres of blood within 24 hours of delivery—or more than 1000 millilitres after caesarean delivery—postpartum haemorrhage is diagnosed

If uterine atony is suspected (a soft, boggy uterus), perform bimanual uterine massage and administer intramuscular oxytocin 10 international units. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

Pathway differentiation is driven by explicit trigger statements in the source guideline. Key branch rules include: If uterine atony is suspected (a soft, boggy uterus), perform bimanual uterine massage and administer intramuscular oxytocin 10 international units. If bleeding persists, give intravenous ergometrine 0.2 milligrams (unless contraindicated by hypertension) or intravenous carboprost 250 micrograms. When genital tract trauma is identified, repair lacerations under adequate analgesia or anaesthesia. When retained placental tissue is suspected, perform manual removal of placenta under appropriate analgesia and antibiotic prophylaxis. If shock develops—defined as systolic blood pressure below 90 millimetres of mercury, tachycardia exceeding 120 beats per minute, altered mental status, or oliguria—administer 2 litres of intravenous crystalloid over 30 minutes, transfuse packed red blood cells based on haemoglobin and ongoing loss, and activate massive transfusion protocol if indicated. These branch statements should be interpreted sequentially, with immediate stabilization pathways prioritized before definitive or downstream treatment pathways.

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Condition 5: Sepsis

If the patient has uterine tenderness, offensive vaginal discharge, or history of recent instrumentation or abortion, intravenous broad-spectrum antibiotics—ceftriaxone 2 grams and

metronidazole 500 milligrams—are indicated immediately, without waiting for culture results. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

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Condition 6: Coagulopathy

If disseminated intravascular coagulation is suspected—characterised by concurrent thrombocytopenia, elevated D-dimer, prolonged prothrombin time, and low fibrinogen—administer fresh frozen plasma, cryoprecipitate, and platelet transfusion according to local massive transfusion protocol and haematology guidance. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

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Condition 7: Eclampsia and Severe Pre

If systolic blood pressure is 160 millimetres of mercury or higher, or diastolic blood pressure is 110 millimetres of mercury or higher, administer intravenous labetalol or hydralazine to reduce mean arterial pressure gradually—avoiding precipitous drops that may compromise placental perfusion. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

Pathway differentiation is driven by explicit trigger statements in the source guideline. Key branch rules include: If systolic blood pressure is 160 millimetres of mercury or higher, or diastolic blood pressure is 110 millimetres of mercury or higher, administer intravenous labetalol or hydralazine to reduce mean arterial pressure gradually—avoiding precipitous drops that may compromise placental perfusion. When severe features are present—including impaired consciousness, pulmonary oedema, oliguria, thrombocytopenia, elevated transaminases, or renal impairment—urgent delivery is indicated regardless of gestational age. These branch statements should be interpreted sequentially, with immediate stabilization pathways prioritized before definitive or downstream treatment pathways.

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Condition 8: Head Injury When a patient presents with head injury and Glasgow Coma Scale score below 8, secure the airway with endotracheal intubation, prevent hypoxia and hypotension, administer intravenous mannitol or hypertonic saline only if signs of cerebral herniation are present, and arrange urgent non

If Glasgow Coma Scale score is between 8 and 12, admit for observation, repeat neurological assessments every 30 minutes initially, and obtain urgent neuroimaging and neurosurgical consultation. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

Pathway differentiation is driven by explicit trigger statements in the source guideline. Key branch rules include: If Glasgow Coma Scale score is between 8 and 12, admit for observation, repeat neurological assessments every 30 minutes initially, and obtain urgent neuroimaging and neurosurgical consultation. When Glasgow Coma Scale score is 13 or higher but red flags are present—including vomiting, convulsions, confusion, cerebrospinal fluid leak, anisocoria, or

lateralising neurological signs—admit for 24-hour observation and serial assessments. These branch statements should be interpreted sequentially, with immediate stabilization pathways prioritized before definitive or downstream treatment pathways.

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Condition 9: Abdominal Injury When a patient presents with blunt abdominal trauma and signs of shock—tachycardia above 120 beats per minute, systolic blood pressure below 90 millimetres of mercury, altered mental status, or poor capillary refill—initiate immediate intravenous resuscitation with two large

If penetrating abdominal trauma is present with an impaled object, do not remove it—stabilise the object with dressings and immobilise the patient during transport. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

Pathway differentiation is driven by explicit trigger statements in the source guideline. Key branch rules include: If penetrating abdominal trauma is present with an impaled object, do not remove it—stabilise the object with dressings and immobilise the patient during transport. If no impaled object but external bleeding is present, apply direct pressure with sterile gauze and administer intravenous broad-spectrum antibiotics per local trauma protocol. When abdominal pain persists or examination reveals rigidity, rebound tenderness, or absent bowel sounds in a stable patient, admit for serial abdominal examinations, repeat FAST, and surgical review within 6 hours. These branch statements should be interpreted sequentially, with immediate stabilization pathways prioritized before definitive or downstream treatment pathways.

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Condition 10: Wheeze and Acute Asthma Exacerbation When a patient presents with wheeze and respiratory distress, stratify severity by ability to speak, respiratory rate, accessory muscle use, oxygen saturation, and mental status

If the patient is in severe exacerbation—unable to complete sentences, respiratory rate above 30

breaths per minute, oxygen saturation below 92% on room air, or pulsus paradoxus above 25 millimetres of mercury—administer high-dose inhaled salbutamol via oxygen-driven nebuliser, intravenous magnesium sulfate 1.2–2 grams over 20 minutes, and oral or intravenous corticosteroids. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

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Condition 11: Fever in Children and Adults When a child older than 60 days presents with fever lasting less than 7 days and no danger signs, assess hydration, malaria risk, respiratory symptoms, urinary symptoms, rash, neck stiffness, and immunosuppression

If complete blood count is normal and malaria antigen test is negative, consider viral or atypical infection and offer outpatient management with antipyretics, oral rehydration, and clear safety-netting instructions—including return if fever persists beyond 5 days, develops rash, or shows signs of dehydration or lethargy. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

Pathway differentiation is driven by explicit trigger statements in the source guideline. Key branch rules include: If complete blood count is normal and malaria antigen test is negative, consider viral or atypical infection and offer outpatient management with antipyretics, oral rehydration, and clear safety-netting instructions—including return if fever persists beyond 5 days, develops rash, or shows signs of dehydration or lethargy. If malaria test is positive and the patient is stable with oral intake, treat as outpatient with weight-based artemisinin-based combination therapy per national policy. If severe malaria is suspected—impaired consciousness, repeated seizures, shock, severe anaemia, or acidosis—admit immediately, start parenteral artesunate, monitor glucose and renal function, and request urgent senior review. These branch statements should be interpreted sequentially, with immediate stabilization pathways prioritized before definitive or downstream treatment pathways.

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Condition 12: Hyperkalaemia When serum potassium exceeds 5

If serum potassium exceeds 6.0 millimoles per litre without ECG changes, shift potassium into cells using intravenous insulin 10 units with 50 millilitres of 50% dextrose, nebulised salbutamol 10 puffs, and intravenous sodium bicarbonate 50 millilitres of 8.4% if metabolic acidosis is confirmed. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

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Condition 13: Diabetic Ketoacidosis When a patient presents with polyuria, polydipsia, nausea, vomiting, abdominal pain, Kussmaul breathing, fruity breath odour, blood glucose above 11 millimoles per litre, ketonaemia or ketonuria, and arterial pH below 7

If shock is present—systolic blood pressure below 90 millimetres of mercury, tachycardia above 120 beats per minute, or altered mental status—administer 1–2 litres of 0.9% sodium chloride over 30–60 minutes before starting insulin infusion. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

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Condition 14: Burns When a patient presents with partial

If facial burns are present, assess airway patency urgently—consider early intubation if stridor, hoarseness, or soot in sputum is noted. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

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Condition 15: Upper Gastrointestinal Bleeding When a patient presents with haematemesis, melaena, or haematochezia and signs of shock—systolic blood pressure below 90 millimetres of mercury, tachycardia above 120 beats per minute, or altered mental status—secure two large

If haemoglobin is below 10 grams per decilitre and bleeding is active, transfuse packed red blood cells to maintain haemoglobin above 7–8 grams per decilitre in most patients—or above 9–10 grams per decilitre in those with cardiovascular disease. In clinical practice, this condition should be approached as a structured pathway problem: the clinician first defines urgency and physiological

stability, then identifies discriminating findings that determine which branch of management is appropriate for the current presentation.

Pathway differentiation is driven by explicit trigger statements in the source guideline. Key branch rules include: If haemoglobin is below 10 grams per decilitre and bleeding is active, transfuse packed red blood cells to maintain haemoglobin above 7–8 grams per decilitre in most patients—or above 9–10 grams per decilitre in those with cardiovascular disease. When tranexamic acid is used, administer intravenous loading dose of 1 gram over 10 minutes, followed by 1 gram infusion over 8 hours—only if endoscopy can be performed within 8 hours and bleeding is not due to varices or malignancy. If *Helicobacter pylori* testing is positive, initiate eradication therapy with proton pump inhibitor, amoxicillin, and clarithromycin per local antimicrobial guidelines—not ranitidine. These branch statements should be interpreted sequentially, with immediate stabilization pathways prioritized before definitive or downstream treatment pathways.

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